Physics-Posterior Distillation: Narrowing the Amortization Gap in Calibrated Sparse-Sensor Source Localization

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Abstract

Amortized neural solvers for Bayesian inverse problems can be fast and, with conformal calibration, provably well-covered — yet calibration says nothing about whether their uncertainty is unnecessarily broad. We study this *amortization gap* in sparse-sensor pollution-source localization, where a tractable advection–diffusion simulator lets us compute the exact Bayesian posterior of every training and evaluation scenario. These exact posteriors serve twice: as dense, physics-derived *supervision*, and as the *reference* that measures how much of the Bayesian information limit a learned solver recovers. Standard location-label training produces a conformally calibrated DeepSets localizer whose 90% regions are nevertheless a median $5.6\times$ larger than the exact-posterior regions, and which reacts to added sensors at only 47% of the Bayes-optimal rate. Distilling mildly tempered exact posteriors into the *identical* architecture — changing only the training target — reduces the median inefficiency to $3.9\times$ and raises the sensor-response ratio to 0.87; raw, untempered posterior targets fail to optimize. Point accuracy barely changes throughout: the deficiency was posterior approximation, not localization. A data-scaling study shows both supervisions follow parallel power laws with no plateau by 40k scenarios — physics supervision acts as a $\sim 2\times$ data multiplier ($p = 5.5 \times 10^{-13}$, paired) — and a stratified audit localizes the gap to information-rich scenarios, where the model rather than the sensor network is the binding constraint. The recipe replicates on a second, methane-configured Gaussian-plume testbed (3-D dispersion, stability classes, kg/h release rates): on identifiable scenarios, inefficiency falls from $32.5\times$ to $8.5\times$. All coverage claims are conformally verified at matched 90% coverage, separating calibration from sharpness. The simulator, frozen scenarios, exact per-scenario posteriors, and full pipeline are released.

1 Introduction

Neural amortized inference has made Bayesian inverse problems fast: a network trained on simulator draws maps observations to a posterior estimate in a millisecond, and split conformal prediction can wrap that estimate into regions with finite-sample coverage guarantees. For decision-relevant inverse problems — our running example is locating a pollutant release from a handful of fixed sensors, a task with growing regulatory stakes [U.S. EPA, 2024; European Union, 2024; UNEP, 2024] — this combination looks like everything one could ask for: speed and provable calibration. It is not. Coverage is a statement about *frequency*, not *efficiency*: a 90% region that covers 90% of the time may still be far larger than the observations warrant, and a localizer that is calibrated but vague fails the responder who must decide where to send a crew.

The obstacle to doing better is the supervision itself. Standard training data provide only the *realized* parameter — "the source for this example was here" — so the network is never shown what its uncertainty *should* look like: which alternative locations also explain the readings, when the posterior should form a ridge or split into modes, how much an additional sensor should tighten the answer. Posterior shape must be inferred statistically across many one-label examples, and the same limitation afflicts evaluation: without a reference posterior, one cannot say whether a calibrated region is efficient or bloated. **Our core observation is that when training data come from a tractable simulator — as they do throughout simulation-based inference — the complete Bayesian posterior of every training scenario can be computed offline and used both ways: as a dense, physics-derived teacher, and as the reference standard that measures how much of the Bayesian information limit the amortized solver recovers.**

Contributions. **(1) Physics-posterior distillation for amortized inverse problems.** We formulate training against exact simulator posteriors as soft targets (Sec. 4.2), symmetry-consistently augmented, and show that changing *only* the supervision — same architecture, data, optimizer, and inference cost — substantially improves uncertainty quality: median calibrated-region inefficiency $5.6\times \to 3.9\times$, sensor-response ratio $0.47 \to 0.87$, with point accuracy nearly unchanged. We further identify a necessary ingredient: *raw* exact-posterior targets fail to optimize; mild Gaussian tempering of the teacher is what makes the physics signal learnable. **(2) An exact-posterior audit of the amortization gap.** Applying the identical conformal procedure to learned and exact posteriors places both at verified 90% coverage, so region-size ratios isolate sharpness from calibration; a controlled dose–response protocol measures whether learned uncertainty reacts to added sensors, degraded geometry, and noise in Bayes-proportional fashion (Sec. 4.4). The enabling infrastructure — simulator, frozen splits, per-scenario exact posteriors, and a preregistered evaluation protocol with validation gates — is released as a public benchmark.

Findings at a glance. (i) Point-label supervision yields calibration without efficiency: $5.6\times$ median inefficiency and half-strength information response (Sec. 5.1). (ii) Fixed tempered physics-posterior distillation recovers a third of the sharpness gap and most of the sensor-response gap; raw teachers are an optimization trap (Sec. 5.2). (iii) The gap is data-limited, not a capacity floor: both supervisions scale as $\approx \text{size}^{-0.46}$ with no plateau by 40k scenarios, and distillation is a constant $\sim 2\times$ data multiplier; stratification shows the gap lives where data are richest, and that with sparse sensors the network — not the model — is the binding constraint (Sec. 5.3). (iv) The recipe transfers: on a methane-configured 3-D Gaussian-plume testbed (CH4-500), distillation cuts identifiable-scenario inefficiency $32.5\times \to 8.5\times$, sharper on 94% of scenarios (Sec. 5.4). (v) Portable implementation lessons: the standard conformal score silently saturates in floating point on sharp probability maps; fixed-node rate quadrature cannot marginalize peaked likelihoods; the specified architecture memorizes without symmetry augmentation (Secs. 4.3–4.4, 4.1).

2 Related Work

Simulation-based inference and posterior distillation. Amortized neural posterior estimation trains on simulator draws and is the standard route to fast Bayesian inversion. Closest to our method, Distill-Belief [Shi et al., 2026] distills a Bayes-correct particle-filter teacher into a student for closed-loop source localization; we distill *exact grid posteriors* with an explicit tempering/annealing treatment and measure the result against the same posteriors at matched conformal coverage. CP4SBI [Cabezas et al., 2025] conformally calibrates SBI credible sets but does not use posteriors as supervision nor audit sharpness against an exact reference.

Conformal prediction for localization and inverse problems. Conformal methods provide distribution-free coverage for acoustic localization [Rozenfeld and Laufer-Goldshtein, 2025], imaging inverse problems [Wen et al., 2024], network source detection [Jian et al., 2026], and anatomical landmarks [Jonkers et al., 2025]. We use conformal prediction not as a contribution but as a *control*: it equalizes coverage across models so that region size becomes a pure sharpness measurement.

Source term estimation. Atmospheric source term estimation is reviewed by Hutchinson et al. [2017]; Bayesian/MCMC plume inversion [Newman et al., 2024] is rigorous but per-scenario and slow; physics-informed neural networks recover advection–diffusion sources [Chuprov et al., 2025;

Table 1: Data-generation prior (sampled independently per scenario; LogUniform(a, b) is uniform in log-space).

Source location	$\boldsymbol{x}_s$	$\mathrm{U}([0.1, 0.9]^2)$	Sensor count	N	U int $\{4..12\}$
Release rate	q	$\mathrm{LogU}(0.5, 5)$	Sensor positions	$\boldsymbol{x}_i$	i.i.d. $\mathrm{U}([0, 1]^2)$
Wind speed	$\lVert\boldsymbol{u}\rVert$	$\mathrm{U}(0.5, 2.0)$	Noise std	σ	$\mathrm{LogU}(0.01, 0.10)$
Wind direction	θ_u	$\mathrm{U}(0, 2\pi)$	Reading dropout	p_{drop}	$\mathrm{U}(0, 0.20)$
Diffusivity	D	$\mathrm{LogU}(0.002, 0.02)$	Observation times	t_k	$k/30$, $k \leq 30$

Anague et al., 2025] without finite-sample coverage. "Information density" in inverse problems [Bangerth et al., 2022] formalizes recoverability; our exact-posterior audit operationalizes a closely related question — how much of the recoverable information an amortized solver uses.

3 Setup: Simulator, Observations, and Exact Posteriors

A scalar concentration on $\Omega = [0, 1]^2$ obeys the constant-coefficient advection–diffusion equation with wind $\boldsymbol{u}$, diffusivity D, and a continuous point release of rate q at $\boldsymbol{x}_s$ regularized by a blob of width $\sigma_s = 1/128$; the unit-rate sensor response has the closed Green's-function form

$$A(\boldsymbol{x}, t; \boldsymbol{x}_s, \boldsymbol{u}, D) = \int_0^t \frac{1}{\pi\, w_s} \exp\left(-\frac{\lVert \boldsymbol{x} - \boldsymbol{x}_s - \boldsymbol{u}\, s\rVert^2}{w_s}\right) \mathrm{d}s, \qquad w_s = 4Ds + 2\sigma_s^2, \tag{1}$$

evaluated by 64-node Gauss–Legendre quadrature after a log-age substitution (validated in Appendix A). Sensor i takes 30 noisy normalized readings $y_{ik} = qA(\boldsymbol{x}_i, t_k)/C_{\text{ref}} + \varepsilon_{ik}$, $\varepsilon_{ik} \sim \mathcal{N}(0, \sigma^2)$, with Bernoulli reading dropout (dropped readings omitted, never imputed). Scenario parameters are drawn from the prior of Table 1; splits (10k/1k/2k/2k) are by complete scenario. A gallery of scenarios and benchmark statistics are in Appendix G (Figures 10–11).

Physical correspondence. The setup is dimensionless, but its parameter ranges are dimensionally forced onto physically sensible scales by choosing a domain length and observation window. Under $L = 500$ m (a facility-scale site) and $T = 5$ min, the wind prior corresponds to 0.8–3.3 m/s (light-to-moderate winds), the diffusivity prior to horizontal eddy diffusivities of 1.7–17 m^2/s, readings arrive every 10 s, grid cells are 7.8 m, and the source blob is a $\sim$4 m piece of equipment. Because readings are reference-normalized, release rate and noise map to signal-to-noise ratios rather than absolute emission rates. The idealization — 2-D, constant known meteorology, no obstacles or background — is deliberate: it is what makes the exact posterior computable, so the uncertainty machinery can be validated against ground truth before confronting real atmospheric complexity (Sec. 6).

Exact posteriors. For candidate cell c of a 64×64 grid with unit-rate response vector $\boldsymbol{g}_c$ at the (known) meteorology, $p(\boldsymbol{y} \mid c, q) = \mathcal{N}(\boldsymbol{y}; q\boldsymbol{g}_c, \sigma^2\mathbf{I})$; marginalizing q over its LogUniform prior and normalizing over cells yields the exact grid posterior $p(c \mid \boldsymbol{y})$ — exact with respect to the synthetic simulator and its grid discretization, not real atmospheric physics. It uses the true meteorology and noise level: the tightest honest region the data permit. Numerical care is required: the q-likelihood peak can be $\sim 1000\times$ narrower than fixed quadrature node spacing, so we use a peak-aware composite rule validated to $< 10^{-6}$ (Sec. 4.4, Appendix B). Posteriors are computed offline for *all* splits — for train and validation they become teachers (Sec. 4.2); for calibration and test they become the audit reference (Sec. 4.4).

4 Method

4.1 M0: amortized posterior estimation and its supervision gap

The estimator (fixed across all methods) is a permutation-invariant DeepSets network [Zaheer et al., 2017] of 4.87M parameters: Fourier-featurized reading tokens $[\mathrm{Fourier}(x_i), \mathrm{Fourier}(y_i), \mathrm{Fourier}(t_k), \mathrm{asinh}(y_{ik}/\sigma)]$, a shared token MLP, masked mean+max pooling, a meteorology-context MLP, and a decoder producing a softmax over the 64^2 grid. Baseline supervision is the standard one: cross-entropy to a smoothed one-hot target (Gaussian bump, std 0.75 cells) at the realized source; AdamW, cosine schedule, batch 256, early stopping on validation

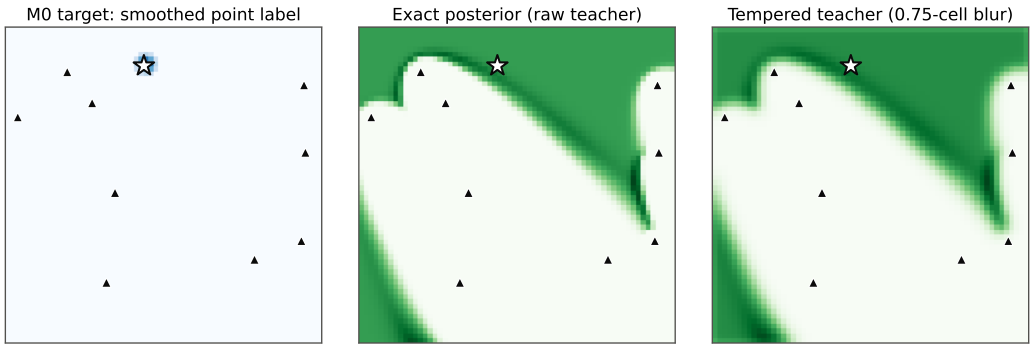

Figure 1: **Three supervision signals** for the same scenario (star = true source; triangles = sensors). Left: the standard smoothed point-label target (M0). Center: the raw exact posterior — Bayes-optimal but an optimization trap. Right: the tempered teacher used by M1 — the same physics shape at a learnable sharpness.

NLL, two-plus seeds. One protocol amendment proved necessary: each batch receives a random D4 symmetry of the square applied jointly to sensors, wind, and source — an exact simulator symmetry under which the prior is invariant — without which this architecture *memorizes* the 10k training scenarios with zero generalization (Appendix C). The supervision gap is structural: the realized location tells the network nothing per-example about ambiguity ridges, multimodality, or how uncertainty should respond to the sensor configuration.

4.2 M1: physics-posterior distillation with a fixed tempered teacher

We replace the target with the scenario's exact posterior, tempered by an isotropic Gaussian of $\sigma_T = 0.75$ cells (matching the baseline bump):

$$\mathcal{L}_{\text{distill}}(\theta) = -\sum_{c} \big[G_{\sigma_T} * p(\cdot \mid \boldsymbol{y})\big](c)\, \log p_\theta(c \mid \boldsymbol{y}), \tag{2}$$

with the teacher grid permuted by the exact same D4 element as the inputs under augmentation (tempering is isotropic, so it commutes with the permutation). Early stopping uses validation teacher cross-entropy — the criterion must match the objective: selecting by true-cell NLL, under which the sharp teacher itself scores poorly, truncates training and *worsens* sharpness ($10.8\times$). Everything else — architecture, data, optimizer, schedule, inference cost — is unchanged from M0. Figure 1 contrasts the three supervision signals on one scenario: the point-label bump carries no ambiguity structure, the raw posterior is too sharp to optimize, and the tempered teacher preserves the physics shape at a learnable scale. Tempering is not cosmetic: with the raw teacher ($\sigma_T = 0$), training fails outright (Sec. 5.2), and a smooth-to-sharp teacher annealing schedule we explored did not improve on the fixed temper (Appendix C).

4.3 Conformal regionization: separating calibration from sharpness

All models (and the exact posteriors themselves) are wrapped by the identical randomized highest-density-mass split conformal procedure: score $s_j = \sum_{c:P_j(c)>P_j(c_j^\star)} P_j(c) + U_j \cdot (\text{tied mass})$ on 2000 calibration scenarios, threshold at the $\lceil (n+1)(1-\alpha) \rceil$-th smallest score, regions filled by descending probability. Under exchangeability every wrapped model enjoys $\Pr(c^\star \in \mathcal{C}_\alpha) \geq 1 - \alpha$ — marginal, in-distribution, not conditional, and not shift-robust. This is the control that makes sharpness measurable: with coverage equalized by construction, region size differences reflect posterior quality alone. One numerical amendment is mandatory on sharp maps: the exceedance mass $1 - \epsilon$ with $\epsilon \sim 10^{-30}$ rounds to exactly 1.0, creating a score atom that silently destroyed coverage (0.84 instead of 0.90 on the exact posterior); we compute throughout in complement (tail-mass) space, which is exact-arithmetic-equivalent and representable at both ends (thresholds reach 4×10^{-199}).

Table 2: Main results (test split, $n = 2000$; coverage target 0.90; area as fraction of the domain; seed 1, spread across three seeds in Table 3). M1 changes only the training target of M0.

	M0 (point labels)	M1 (distilled)	Exact oracle	Peak sensor
Coverage @ 90%	0.875	0.899	0.907	—
CP 95% CI	[.860, .889]	[.884, .911]	[.893, .919]	—
Region area (median)	0.068	0.053	0.013	—
Inefficiency (median)	5.6×	3.9×	1×	—
Area Spearman vs. oracle	0.848	0.893	—	—
JSD to oracle (nats)	0.56	0.50	0	—
Sensor-response ratio	0.47	0.87	1.0	—
MAP error (median)	0.103	0.094	0.023	0.315

4.4 Measurement: the exact-posterior audit

Three preregistered measurements, all on held-out data. **Calibration (H1):** empirical coverage at 90% with exact Clopper–Pearson intervals. **Sharpness (H2):** with learned and exact regions both at verified 90% conformal coverage, the per-scenario area ratio (learned/exact) — the *inefficiency factor* — and the Spearman correlation of areas. An inefficiency of 1 would mean the solver extracts all the location information the data contain; ratios far above 1 quantify the amortization gap (a ratio below 1 is neither expected nor desirable — it would indicate miscalibration absorbed by conformal inflation elsewhere). **Dose–response (H3):** on constructed scenario sets — nested sensor subsets $N = 4..12$ added in fixed order, matched well-spread vs. confined-quadrant geometry pairs, and a noise sweep — the ratio of learned to exact area-response slopes and per-knob rank agreement; region-size language only, since constructed sets are not exchangeable prior draws. Auxiliary: Jensen–Shannon divergence to the exact posterior, MAP error, latency. Validation gates (quadrature convergence, an independent finite-difference solver cross-check, oracle sanity checks, a scenario-leakage audit, and a 96^2 grid-refinement check) are recorded in Appendices A–E; the preregistered-vs-exploratory demarcation and all protocol deviations are listed in Appendix F.

5 Results

5.1 Point-label supervision: calibrated but inefficient

Conformal M0 attains near-nominal coverage — 0.875 $[0.860, 0.889]$ for seed 1 (a shortfall quantified by permutation test as a $p = 0.0045$ finite-sample fluctuation of this split pair; seeds 2–3 cover at 0.896 and 0.912, the exact posterior at 0.907; Appendix D) — and Figure 2 shows the conformal curve tracking the diagonal where raw HPD regions undercover. Calibration, however, conceals inefficiency: at matched 90% coverage M0's regions are a median $5.6\times$ (mean $47.6\times$) larger than the exact-posterior regions, with the worst ratios exactly where the data are most informative (razor-sharp oracle, learned heatmap saturated near 0.4 of the domain; Figure 7, left). Its response to information is directionally correct on all three knobs but weak: 0.47 sensor-count slope ratio, 0.49 noise. Point accuracy is respectable (median MAP error 0.103 vs. oracle 0.023, peak-sensor baseline 0.315): what point labels failed to teach is not *where*, but *how unsure*.

5.2 Fixed-teacher distillation works; the raw teacher does not

Replacing the target with the tempered exact posterior (M1) improves every uncertainty-quality metric while leaving point accuracy nearly unchanged (Table 2): inefficiency $5.6\times \to 3.9\times$, area Spearman $0.848 \to 0.893$, JSD $0.56 \to 0.50$, sensor response $0.47 \to 0.87$, geometry agreement $0.88 \to 0.90$. The contrast that matters methodologically: the *raw* exact-posterior teacher fails outright — median validation HPD area 0.197, worse than point labels — and intermediate mixtures underperform full tempered distillation (validation areas: raw λ=1: 0.197; raw λ=0.5: 0.064; tempered λ=1: 0.055). Near-delta targets provide gradients dominated by a few cells the student cannot yet match; tempering to the scale of the baseline bump makes the same physics signal learnable. Figure 3 shows the effect across the difficulty spectrum: where the oracle is sharp, M0 saturates into a broad blot while M1 concentrates; where the oracle is ridged or multimodal, M1 reproduces the ambiguity structure that M0 only gestures at.

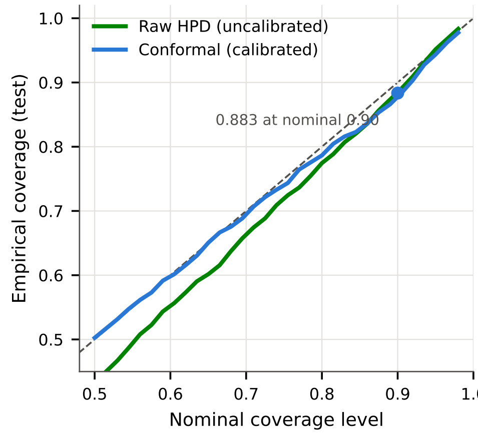

Figure 2: **Coverage vs. nominal** (M0). Conformal regions track the diagonal; raw HPD undercovers. Coverage is thereby equalized across models, making region size a pure sharpness measurement.

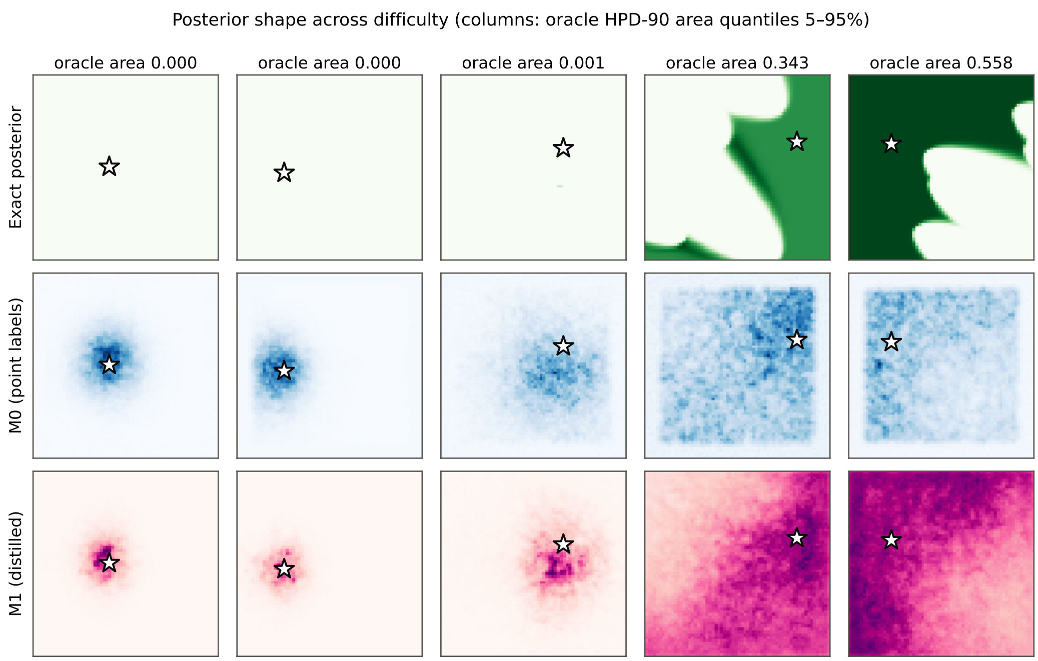

Figure 3: **Posterior shape across difficulty.** Columns: five test scenarios at oracle HPD-90 area quantiles (5–95%). Rows: exact posterior, M0, M1 (same scenario per column; star = true source). Distillation transfers the oracle's shape — sharp peaks, ridges, multimodality — that point labels cannot supervise.

5.3 Significance, and what drives the residual gap

The M0–M1 improvement is statistically unambiguous: a one-sided paired Wilcoxon signed-rank test on per-scenario log inefficiency ratios gives $p = 5.5 \times 10^{-13}$, M1 is sharper on 61% of individual scenarios, and bootstrap 95% CIs on the median inefficiency are $5.6\times$ $[4.7, 6.9]$ (M0) vs. $3.9\times$ $[3.3, 4.8]$ (M1); the ordering is stable across three seeds with no metric overlap (Table 3).

The gap is data-limited, and distillation is worth $\sim$2$\times$ data. Retraining both supervisions at $\{2.5, 5, 10, 40\}$k scenarios (Figure 4) shows both inefficiencies falling as an approximate power law (median inefficiency $\propto$ size$^{-0.46}$) with *no plateau* by 40k — at this architecture the residual gap is

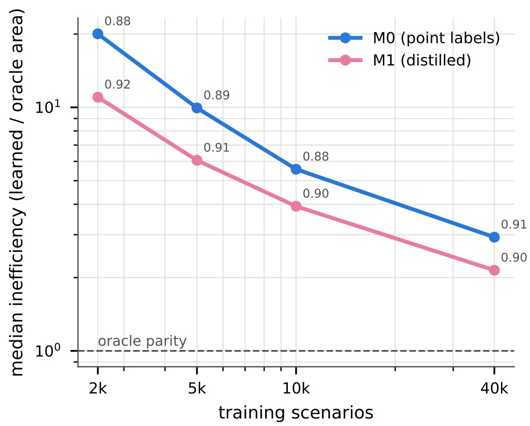

Figure 4: **Data-scaling of the amortization gap** (seed 1; small labels = conformal coverage per point). Both supervisions follow $\approx$ size$^{-0.46}$ power laws with no plateau by 40k; the distilled curve is offset by a constant $\sim 2\times$ in data.

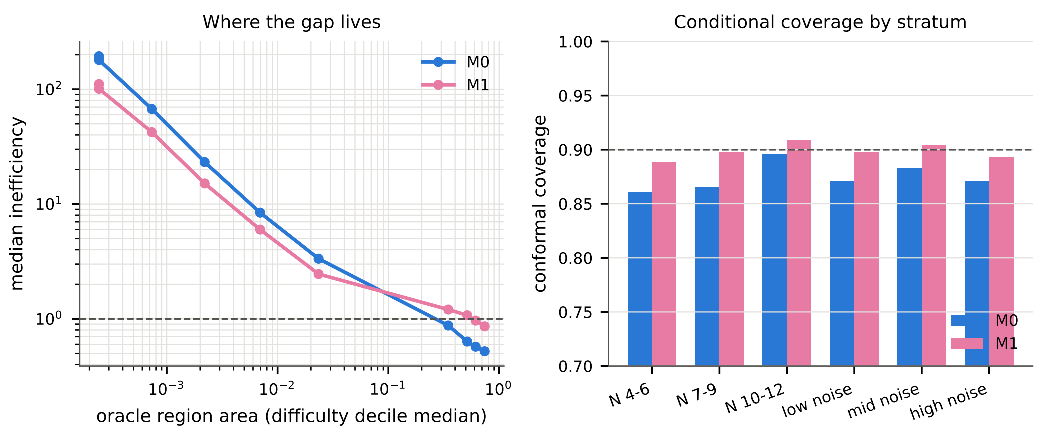

Figure 5: **Stratified audit.** Left: median inefficiency by oracle-difficulty decile — the gap concentrates where the oracle is sharpest. Right: conditional conformal coverage by sensor-count and noise strata; distillation is both sharper and more uniformly calibrated.

predominantly a simulation-budget limit, not a capacity floor. The two curves are near-parallel: M1 at any size matches M0 at roughly *twice* that size (M1@2.5k $\approx$ M0@5k; M1@10k $\approx$ M0@21k, interpolated), so physics-posterior supervision acts as a $\sim 2\times$ data multiplier, constant across the tested range. The best configuration (M1@40k) reaches $2.14\times$ at coverage 0.897.

Where the gap lives. Stratifying the audit (Figure 5) localizes the inefficiency where the *data are richest*: with 4–6 sensors or high noise, even M0 sits near the physical limit (1.2–$1.9\times$) — the sensor network, not the model, is the binding constraint — while with 10–12 sensors or low noise the model is the constraint ($17\times$ for M0, cut to $11\times$ by M1; $194\times \to 111\times$ in the sharpest-oracle decile). Distillation also makes conditional coverage *more uniform*: M1 stays within 0.888–0.909 across all strata where M0 spans 0.861–0.896. (On the broadest-oracle deciles both models' ratios dip below 1; this is the marginal-coverage trade-off — under-covering easy scenarios while over-covering sharp ones — not superiority to Bayes.) Figure 8 shows the mechanism at scenario level: as sensors accrue, the distilled model's mean region-area curve tracks the oracle's closely where M0 under-reacts. The area–error association (Figure 7, right) persists across models, supporting region area as a future abstention signal; per-scenario improvement distributions and remaining knob responses are in Appendix G (Figures 12–13).

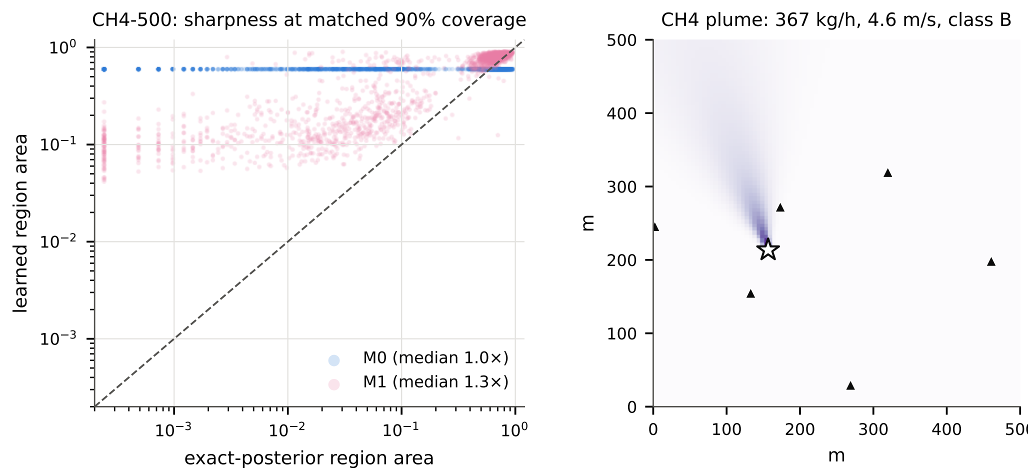

Figure 6: **CH4-500 testbed.** Left: sharpness at matched 90% coverage on methane-configured physics; the identifiable-scenario cloud (small oracle areas) moves toward the diagonal under distillation. Right: an example scenario — a 3-D Gaussian plume in physical units with sensors and true source.

Table 3: Seed stability (three seeds per model): headline metrics as mean [min, max]. Full per-seed values in `ablation_seeds.csv`.

	M0 (point labels)	M1 (distilled)
Coverage @ 90%	0.894 [0.875, 0.912]	0.895 [0.891, 0.899]
Region area (median)	0.072 [0.068, 0.074]	0.053 [0.052, 0.054]
Inefficiency (median)	5.74× [5.57, 5.86]	4.02× [3.92, 4.11]
Area Spearman vs. oracle	0.851 [0.848, 0.854]	0.893 [0.890, 0.897]
JSD to oracle (nats)	0.556 [0.555, 0.557]	0.499 [0.498, 0.500]
MAP error (median)	0.100 [0.098, 0.103]	0.093 [0.092, 0.094]

5.4 CH4-500: the recipe replicates on a methane-configured testbed

To test transfer beyond one simulator, we built a second, methane-configured testbed: a 500×500 m facility site with a steady-state three-dimensional Gaussian plume (ground reflection; Briggs open-country dispersion under Pasquill–Gifford stability classes A–F, known context), release rates 10–500 kg/h spanning the 100 kg/h super-emitter threshold, winds 1–8 m/s, a 3 m release height, 2 m sensor masts, and 0.2–3 ppm sensor noise — the model family of practical methane inversion [Newman et al., 2024]. The plume is closed-form, so the entire pipeline transfers verbatim (implementation gates: exact crosswind symmetry; mass-flux conservation $\iint uC \, \mathrm{d}z \, \mathrm{d}y = Q$ to 4×10^{-6}; 80 ppm at 100 m from a 100 kg/h release in D stability). Conformal coverage holds for all three posteriors (0.894–0.912). The structure of the problem differs sharply from ADV-2D: thin steady plumes miss randomly placed sparse sensors, so 57% of scenarios are weakly identifiable (oracle region $\geq 50\%$ of the site) — itself a network-design finding — and there both models correctly match the near-prior oracle ($\approx 1\times$). On the identifiable third (oracle $< 10\%$ of the site), the main result replicates and amplifies: median inefficiency falls from $32.5\times$ (M0) to $8.5\times$ (M1), with M1 sharper on 94% of scenarios (paired Wilcoxon $p = 1.6 \times 10^{-8}$; means $348 \rightarrow 63$). Under these units, that is the difference between an unactionable region and one approaching survey scale (Figure 6).

6 Discussion, Limitations, and Broader Relevance

Why this is more than a benchmark. The exact posteriors are not merely an evaluation convenience: the central empirical message is that they constitute a *training resource* that standard pipelines leave unused. Changing only the supervision target — no architecture, capacity, data, or inference-time change — moved every uncertainty-quality metric substantially while barely moving

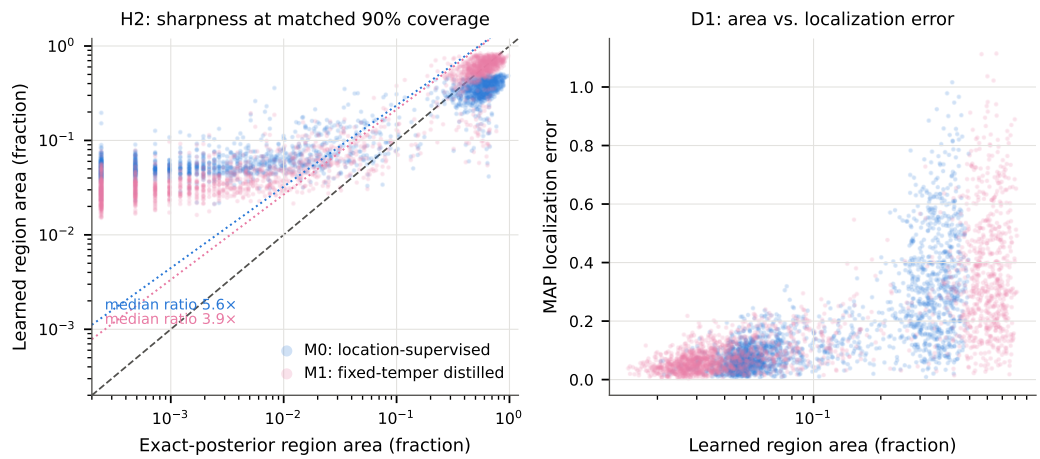

Figure 7: **The amortization gap, measured.** Left: learned vs. exact region area at matched 90% conformal coverage (dotted lines: median ratios). Physics-posterior distillation moves the cloud toward the diagonal. Right: region area vs. MAP error.

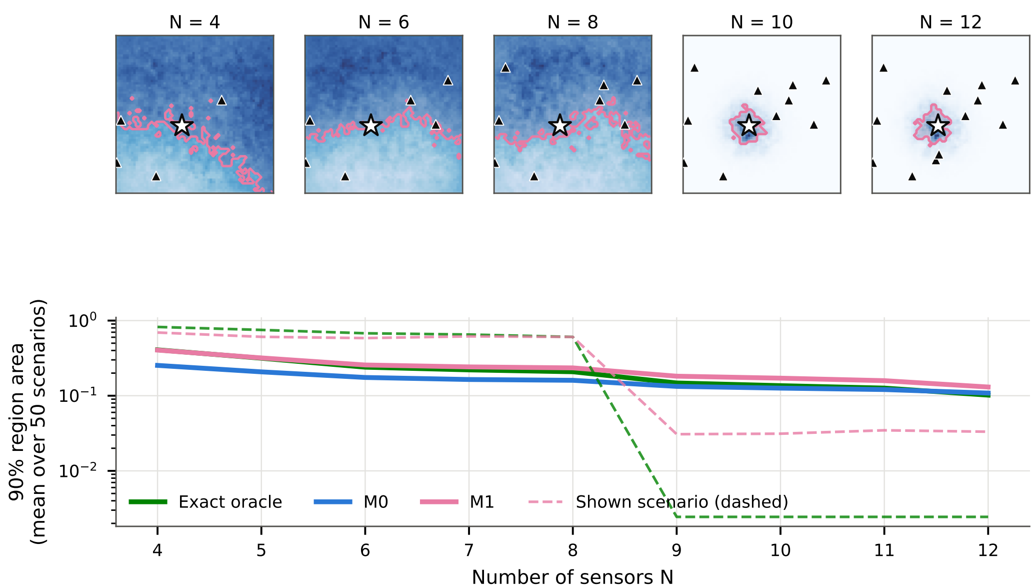

Figure 8: **Information dose–response.** Top: one base scenario as sensors accrue $4 \to 12$ (best model's heatmap; region as translucent fill; star = true source). Bottom: mean 90% region area vs. N for the exact oracle, M0, and M1.

point accuracy, demonstrating that the original failure was posterior approximation, not localization. The benchmark is the instrument that made this effect visible (conformal coverage equalization) and measurable (matched-coverage inefficiency; causal dose–response); the method is what the measurement enabled.

Operational reading. Under the facility-scale correspondence of Sec. 3, the median 90% region shrinks from an equivalent radius of 74 m (M0) to 65 m (M1), with the physical limit at 32 m — movement toward the 50 m radius at which, e.g., U.S. Super-Emitter Program obligations attach to an identified operator [U.S. EPA, 2024]. Region area is also a direct proxy for follow-up survey effort, so the M0 $\to$ M1 improvement corresponds to $\sim$22% less search area per alert at identical confidence; and the stratified audit doubles as network guidance — with 4–6 sensors the network, not the model, binds ($1.2\times$ from the physical limit), while at 10–12 sensors the model binds and training quality dominates. We state plainly that these are readings of a synthetic testbed, not deployment claims.

Positioning. Within simulation-based inference, the recipe is general: wherever training scenarios come from a simulator with a tractable (or accurately approximable) likelihood, per-scenario posterior targets can be computed offline and distilled — our teacher is exact physics rather than a learned network [cf. Shi et al., 2026]. Within scientific ML, it is a physics-*derived* supervision signal, distinct from PDE-residual (PINN-style) losses: the physics enters through the target, and deployment cost is untouched. Within conformal UQ, we use calibration as an experimental control rather than a contribution, which we believe is the right division of labor: conformal guarantees what it can (coverage), and supervision quality determines what it cannot (sharpness).

Limitations. The study is synthetic and two-dimensional, with a single stationary source, constant known meteorology, Gaussian noise of known level, and no obstacles, turbulence, or background; "exact" means exact for this simulator and grid discretization, not for real atmospheric physics. Coverage guarantees are marginal and in-distribution; nothing here addresses distribution shift or field deployment. Distillation requires tractable per-scenario posteriors at training time — automatic in both testbeds here, an approximation problem in messier simulators. The CH4-500 configuration remains idealized (steady-state plume, flat terrain, known stability, fixed release height). Point accuracy is essentially unchanged by the method — it improves uncertainty, not localization. We claim no new conformal algorithm and no new architecture.

7 Conclusion

Exact simulator posteriors can serve as dense physics-derived supervision that improves the sharpness and information sensitivity of amortized Bayesian inverse solvers without changing their architecture or inference-time cost. On a sparse-sensor source localization benchmark built to measure exactly this — conformal coverage equalization plus an exact-posterior reference — point-label training left a $5.6\times$ sharpness gap and half-strength information response; tempered physics-posterior distillation reduced the gap to $3.9\times$ and restored 87% of the Bayes-proportional sensor response. The residual gap is a measured, data-limited property — it follows a power law in simulation budget, with distillation halving the budget required — and the released benchmark makes the supervision, the measurement, and the scaling protocol available to any simulator-based inverse problem.

Reproducibility Statement

A single configuration file drives every stage; one script regenerates the study from a clean checkout; scenario regeneration is bit-identical (SHA-256 verified). We release the simulator and generator, frozen scenario IDs, exact per-scenario posteriors for all splits, dose–response sets, checkpoints for all models and seeds, and the audit pipeline. Total compute: ≈ 2 GPU-hours on one NVIDIA L40S; 300 MB of data.

References

A. N. Angelopoulos and S. Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv:2107.07511*, 2021. Published as *Foundations and Trends in Machine Learning* 16(4):494–591, 2023.

B. Anague, B. Hosseini, I. Karambal, and J. M. Ngnotchouye. Physics-informed neural networks for joint source and parameter estimation in advection–diffusion equations. *arXiv:2512.07755*, 2025.

W. Bangerth, C. R. Johnson, D. K. Njeru, and B. van Bloemen Waanders. Estimating and using information in inverse problems. *arXiv:2208.09095*, 2022 (rev. 2025).

L. M. C. Cabezas, V. S. Santos, T. R. Ramos, P. L. C. Rodrigues, and R. Izbicki. CP4SBI: Local conformal calibration of credible sets in simulation-based inference. *arXiv:2508.17077*, 2025. Accepted, *Phil. Trans. R. Soc. A*.

I. Chuprov, D. Derkach, D. Efremenko, and A. Kychkin. Application of physics-informed neural networks for solving the inverse advection–diffusion problem to localize pollution sources. *arXiv:2503.18849*, 2025.

C. Edwards and R. C. Smith. Rapid parameter inference with uncertainty quantification for a radiological plume source identification problem. *arXiv:2502.17492*, 2025.

European Union. Regulation (EU) 2024/1787 on the reduction of methane emissions in the energy sector. *OJ L, 2024/1787*, 15 July 2024.

M. Hutchinson, H. Oh, and W.-H. Chen. A review of source term estimation methods for atmospheric dispersion events using static or mobile sensors. *Information Fusion*, 36:130–148, 2017.

X. Jian, P. Zhang, L. Tian, F. Ji, W. Liang, W. P. Tay, et al. Conformal prediction for multi-source detection on a network. *Proc. AAAI*, 40(43):36663–36670, 2026. arXiv:2511.08867.

J. Jonkers, F. Coopman, L. Duchateau, G. Van Wallendael, and S. Van Hoecke. Reliable uncertainty quantification for 2D/3D anatomical landmark localization using multi-output conformal prediction. *arXiv:2503.14106*, 2025. *Medical Image Analysis* 110:103953, 2026.

T. Newman, C. Nemeth, M. Jones, and P. Jonathan. Probabilistic inversion modeling of gas emissions: a gradient-based MCMC estimation of Gaussian plume parameters. *arXiv:2408.01298*, 2024.

V. Rozenfeld and B. Laufer-Goldshtein. Conformal prediction for manifold-based source localization with Gaussian processes. In *ICASSP*, 2025. arXiv:2409.11804.

Y. Shi, Z. Song, M. Yang, C. Liu, and W. Liu. Distill-Belief: Closed-loop inverse source localization and characterization in physical fields. *arXiv:2604.26095*, 2026.

UN Environment Programme (IMEO). Methane Alert and Response System (MARS). `https://www.unep.org/topics/energy/methane/methane-alert-and-response-system-mars`; 1,200 notifications over first two years (Baku press release, 15 Nov. 2024).

U.S. Environmental Protection Agency. NSPS OOOOb / EG OOOOc; Super-Emitter Program. *89 FR 16820*, March 8, 2024 (40 CFR Part 60).

J. Wen, R. Ahmad, and P. Schniter. Task-driven uncertainty quantification in inverse problems via conformal prediction. In *ECCV*, LNCS 15118, pp. 182–199, 2024. arXiv:2405.18527.

M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Poczos, R. Salakhutdinov, and A. Smola. Deep sets. In *NeurIPS*, 2017. arXiv:1703.06114.

A Quadrature and Physics Validation

Age-integral quadrature: 64-node Gauss–Legendre under $s = te^{-\tau}$, $\tau_{\max} = 12$ (tuned; at $\tau_{\max} = 40$ the gate fails at 2×10^{-4}). 64- vs 128-node median relative error 3.6×10^{-11} (gate $< 10^{-6}$). Zero-wind closed form (exponential integrals): median relative error 1.2×10^{-13}. Independent second-order finite-difference solver (central differences, Heun, CFL-stable, $h = 1/768$, enlarged Dirichlet domain), 20 random scenarios: max 0.50% relative L_2 at sensors (median 0.06%) over the 17 scenarios with signal; the 3 no-signal scenarios agree in absolute RMS to $\leq 8.8 \times 10^{-11}$ (gated at 10^{-4}, 1% of the smallest noise std). Symmetry equivariance 1.7×10^{-13}; linearity 2.7×10^{-16}; scenario-ID leakage audit: zero overlap, zero duplicated content.

B Exact-Posterior Numerics and Sanity Checks

Rate marginalization uses a peak-aware composite Gauss–Legendre rule (panels split at $\hat{q}_c \pm 8\sigma/\|\boldsymbol{g}_c\|$; 32 peak + 2×16 outer nodes, `float64` log-space), validated against a $10\times$-node deep rule and a 200,000-node trapezoid to $\leq 9.4 \times 10^{-7}$ in log-posterior; the preregistered fixed 32-node rule errs by many nats. Normalization within 3.4×10^{-13}. Mode $\to$ true cell as $\sigma \to 0$: 24/24 (sanity sources at cell centers; identifiability conditioning; $\sigma = 10^{-6}$). Confined layouts yield broad posteriors (90% HPD area 0.48 vs. 0.064 well-spread). Conformal-on-exact-posterior: 0.907 $[0.893, 0.919]$. Area contracts with q (87.5% of scenarios, mean $\Delta = -0.005$) and grows with σ (100%, $+0.024$).

C Training Details and the Memorization Finding

Baseline hyperparameters in Sec. 4.1; 0.011 GPU-h per seed (L40S). Without D4 augmentation the preregistered recipe memorizes: train NLL $\to 1.5$ while validation NLL never improves below the uniform level (≈ 7.96; $\log 4096 = 8.32$) and validation MAP error (0.44) is worse than the peak-sensor heuristic (0.37); verified not attributable to mixed precision, learning rate, or the context branch (Figure 9).

M1: early stop on validation teacher-CE; winner selected on validation among $\{\lambda, \text{temper}\}$ variants (Sec. 5.2). A progressive smooth-to-sharp teacher annealing schedule (hold $\sigma_T = 0.75$, linear anneal

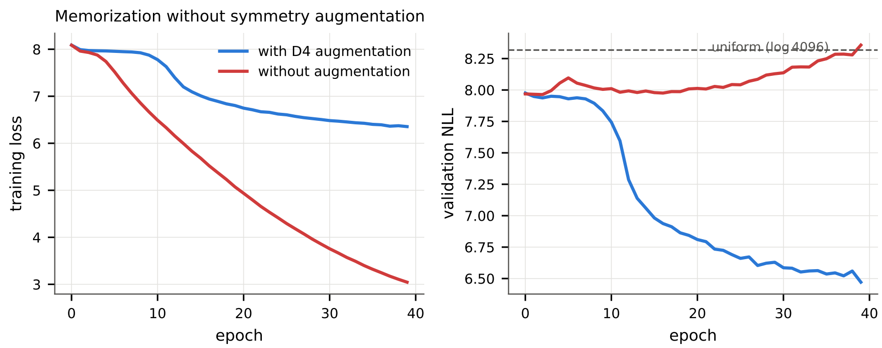

Figure 9: **Memorization without symmetry augmentation.** Training loss falls either way (left), but without the exact D4 augmentation validation NLL never leaves the uniform level (right).

to $\sigma_{\text{end}} \in \{0.30, 0.15, 0\}$) was explored with validation-only selection and did not improve on the fixed temper (best curriculum validation HPD-90 median area 0.061 vs. 0.055 fixed; test inefficiency $4.8\times$ vs. $3.9\times$), with sharper end-targets monotonically worse; it is therefore omitted from the main method. Selection records in the released artifact.

D Calibration Detail (H1) and Auxiliary Metrics

M0 seed-1 coverage 0.875 $[0.860, 0.889]$ misses 0.90; investigation: in-sample calibration exact (0.9005), coverage invariant to score randomization (< 0.0005 over 20 draws), calib/test scores indistinguishable (KS $p = 0.51$), split-swap *over*-covers (0.917), permutation $p = 0.0045$ — a finite-sample fluctuation of this (model, split) pair. Coverage at $80/90/95\%$: M0 $0.787/0.875/0.944$; M1 $0.799/0.899/0.949$. Latency 0.48 ms/scenario (batch 1). MAP percentiles (M0): $0.10/0.43/0.78$.

E Grid Refinement

Recomputing exact posteriors at 96^2 on the full test split changes conformal coverage by -0.007 and median area by -0.006 — the 64^2 grid is adequate (the preregistered 100-scenario version cannot resolve the effect: 100-scenario coverage windows span 0.79–0.97).

F Preregistration and Deviations

The three-way audit (H1/H2/H3), the five headline metrics, splits, and gates were preregistered; M1 and M2 are exploratory additions with validation-only selection, test evaluated once per final variant. Protocol deviations, each forced by a correctness failure or ill-posedness caught by the pipeline's own blocking gates: (i) D4 augmentation (Sec. 4.1); (ii) tail-space conformal scores (Sec. 4.3); (iii) peak-aware rate quadrature (Appendix B); (iv) well-posed sanity formulations (Appendix B); (v) full-split grid refinement (Appendix E); (vi) absolute-error FD gating at zero signal (Appendix A). No preregistered hypothesis, metric, or split was changed after seeing data.

G Benchmark Gallery and Additional Visualizations

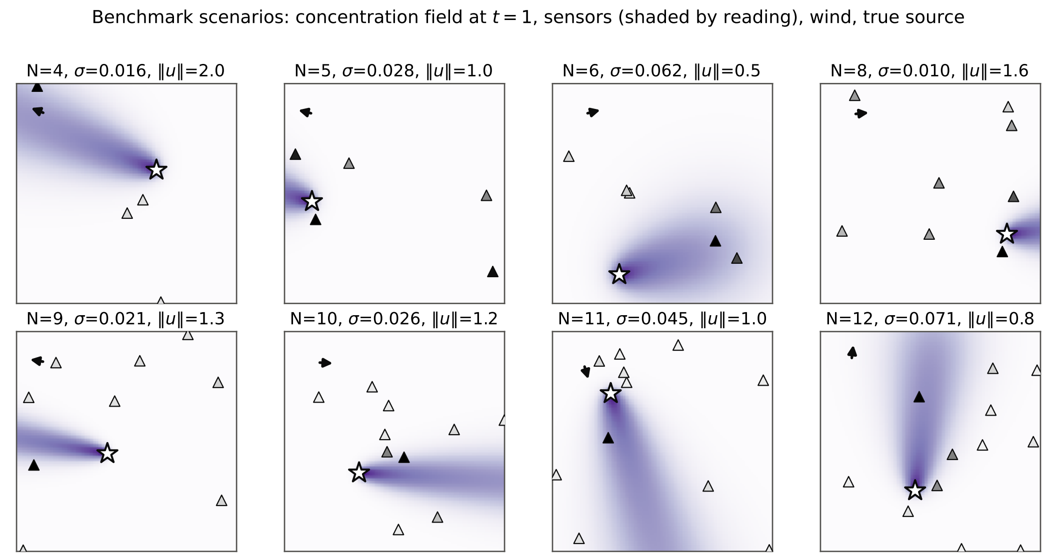

Figure 10: **Benchmark gallery.** Eight test scenarios: concentration field at $t = 1$ (purple), sensors (triangles, shaded by time-averaged reading), wind direction (arrow), true source (star). Sensor count, noise, wind, and geometry vary per the prior of Table 1.

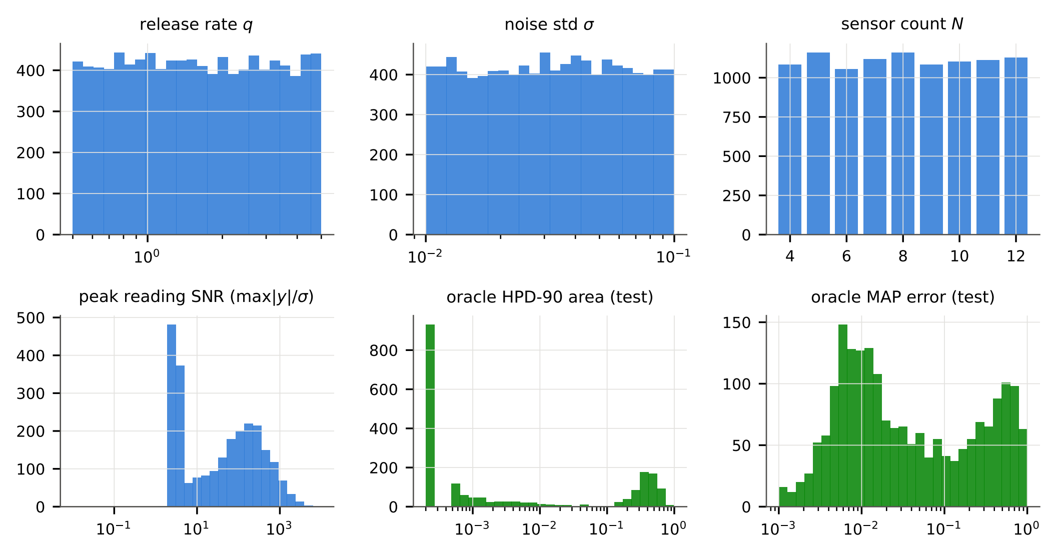

Figure 11: **Benchmark statistics.** Top: prior draws (release rate, noise, sensor count; training split). Bottom: peak reading SNR; oracle HPD-90 area and oracle MAP error on the test split — the benchmark spans effectively unidentifiable to near-exactly identifiable scenarios.

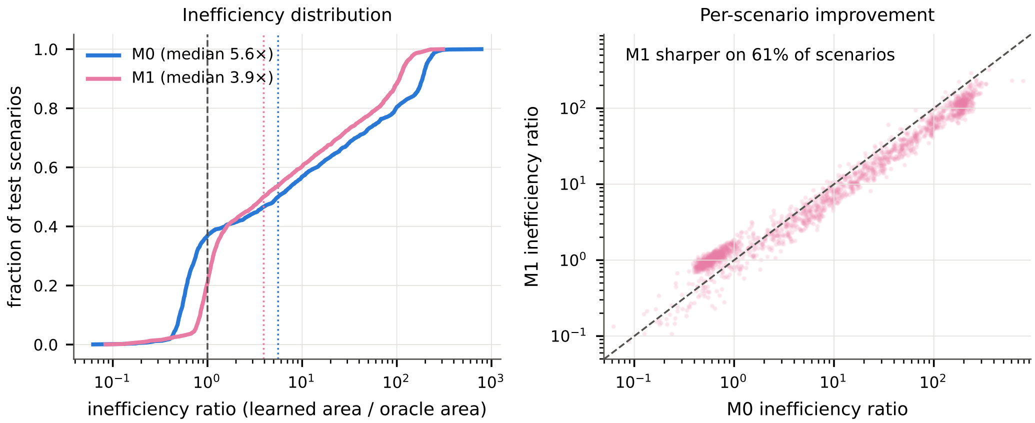

Figure 12: **Per-scenario improvement.** Left: inefficiency ECDFs (dashed: oracle parity). Right: paired per-scenario ratios; points below the diagonal are scenarios where distillation sharpened the region.

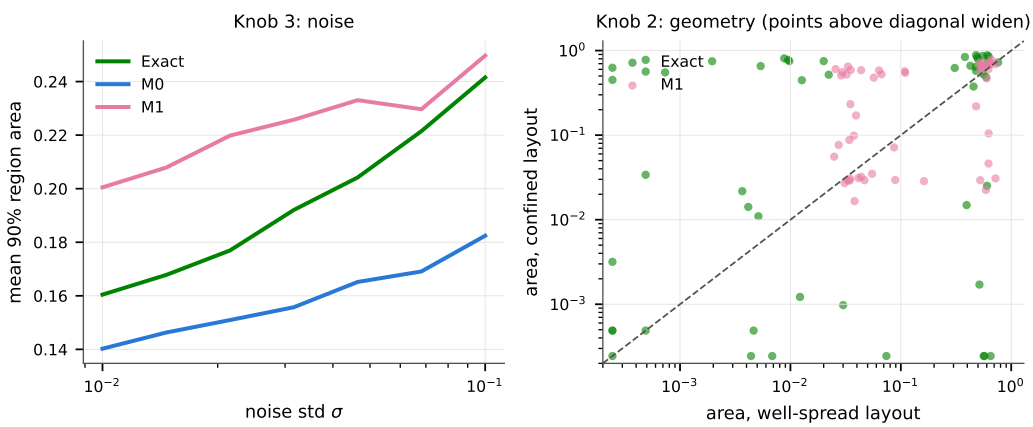

Figure 13: **Remaining dose–response knobs.** Left: mean region area under the noise sweep (Knob 3). Right: matched geometry pairs (Knob 2); points above the diagonal widen under the confined layout, as the physics dictates.

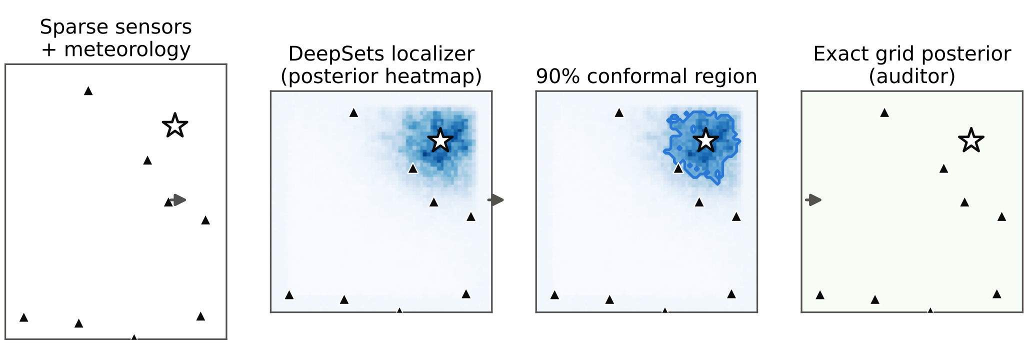

Figure 14: **Pipeline** on a real median-difficulty test scenario (M0; star = true source, triangles = sensors): readings $\rightarrow$ amortized heatmap $\rightarrow$ 90% conformal region; the exact grid posterior is teacher (train/val) and reference (calib/test).

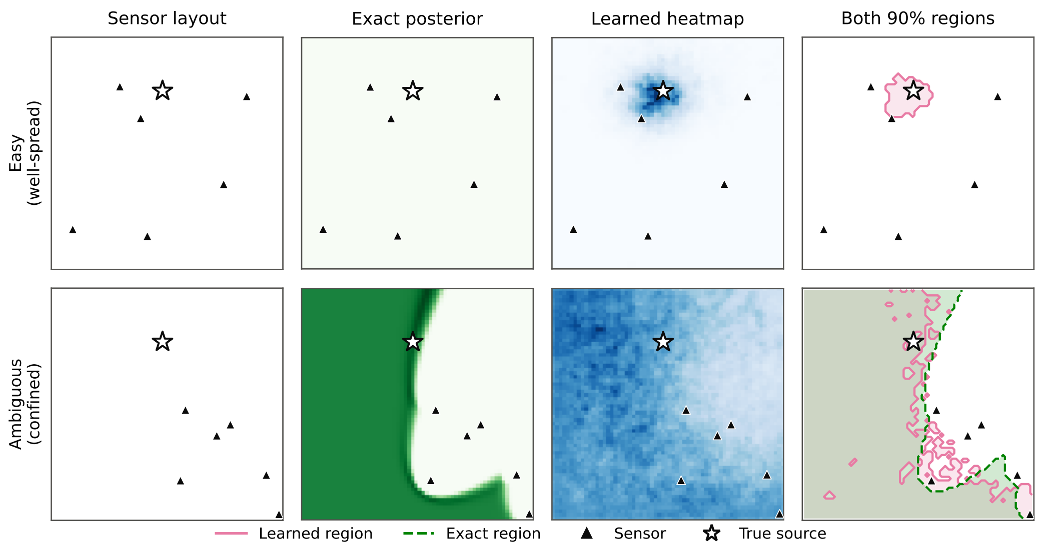

Figure 15: **Easy vs. ambiguous scenario** (heatmaps from the best distilled model): layout, exact posterior, learned heatmap, both 90% regions. The confined layout induces the directional-ambiguity arc; both regions widen accordingly.